The Hardy-range saturation infimum equals one

Candidate full solution packet for arXiv:2601.01821

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Status. Candidate full solution, likely valid, awaiting specialist review. Open Problem 2 of the source is resolved for every stated Hardy exponent: $\varepsilon_p(A; r, R) = 1$ for $0 < p \leq 1$. The source's separate non-radial problem is not addressed.

1 Source problem

Wang studies the anisotropic Hardy space $H^p_A(\mathbb{R}^n)$ associated with an expansive matrix A and the wavelet frame operator

$$U_{\psi,\phi} f = \sum_{j\in\mathbb{Z}} \sum_{k\in\mathbb{Z}^n} \langle f, \phi_{j,k}\rangle \psi_{j,k}.$$

Suppose A^* has a real eigenvalue λ, the generator ψ is radial and satisfies the source's decay, vanishing-moment, and band-limited admissibility conditions, and

$$\operatorname{supp} \widehat{\psi} \subset \mathcal{A}(r, R) = \{\xi : r \leq |\xi| \leq R\}, \qquad |\lambda| > R/r.$$

Let $\mathfrak{R}(A; r, R)$ denote this generator class. The source asks to determine, for $0 < p \leq 1$,

$$\varepsilon_p(A; r, R) := \inf_{\psi\in\mathfrak{R}(A;r,R)} \|U_{\psi,\psi} - \mathrm{Id}\|_{H^p_A \to H^p_A}.$$

Its Theorem 1 proves $\varepsilon_p(A; r, R) \geq 1$, while its Proposition 1 proves equality at the separate Hilbert endpoint $p = 2$ [1].

tion adapted to an anisotropic-annulus support, and asks whether the lower bound 1 persists.

Open Problem 2 (Saturation infimum) Let the hypotheses of Theorem 1 be in force: A is expansive on $\mathbb{R}^n$ and satisfies Assumption 1, A^* admits a real eigenvalue $\lambda \in \mathbb{R}$ with eigenvector $v \in \mathbb{R}^n \backslash \{0\}$, and the spectral threshold $|\lambda| > R/r$ is in force. Denote by $\mathfrak{R}(A; r, R)$ the class of radial generators ψ that satisfy Assumption 1 together with $\operatorname{supp} \widehat{\psi} \subset \mathcal{A}(r, R)$. For each $p \in (0, 1]$, define the saturation infimum

$$\varepsilon_p(A; r, R) \ := \inf_{\psi\in\mathfrak{R}(A;r,R)} \big\|U_{\psi,\psi} - \mathrm{Id}\big\|_{H^p_A \to H^p_A}. \tag{42}$$

Determine the value of $\varepsilon_p(A; r, R)$ across the Hardy range $p \in (0, 1]$.

Theorem 1 delivers the lower bound $\varepsilon_p(A; r, R) \geq 1$, uniformly in $p \in (0, 1]$. The definition (42) extends to the Hilbert endpoint $p = 2$ on identifying $H^2_A(\mathbb{R}^n)$ with $L^2(\mathbb{R}^n)$, and the corresponding endpoint value $\varepsilon_2(A; r, R)$ is supplied by Proposition 1 below.

Proposition 1 (Hilbert-endpoint resolution) *Under the hypotheses of Theorem 1, the saturation infimum at the Hilbert endpoint takes the explicit value*

$$\varepsilon_2(A; r, R) \ = \inf_{\psi\in\mathfrak{R}(A;r,R)} \big\|U_{\psi,\psi} - \mathrm{Id}\big\|_{L^2 \to L^2} \ = \ 1. \tag{43}$$

Figure 1: Open Problem 2 on PDF page 15 of the locally compiled source.

2 Proof intuition

The admissible class is a cone: multiplying a generator by any positive scalar changes neither its support nor its radiality, moments, or decay. At the same time the diagonal frame operator is quadratic in the generator: $U_{t\psi,t\psi} = t^2 U_{\psi,\psi}$. Thus a bounded base frame operator can be damped arbitrarily close to the zero operator.

The distance from the zero operator to the identity is exactly one. Although H^p_A is only quasi-Banach when $p < 1$, its exact p-power triangle inequality is tailor-made for this limit. It bounds the distance from $t^2 U$ to the identity by a quantity converging to one. The source's geometric obstruction supplies the matching lower bound.

This avoids the sharper multiplier estimate contemplated in the source. One does not need to bound an operator norm by the bare L^∞-norm of its symbol; any finite Hardy multiplier bound for one smooth base generator is enough.

3 A general damping lemma

Lemma 1. *Let $0 < p \le 1$, and let X be a quasi-normed vector space satisfying*

$$\|x + y\|_X^p \le \|x\|_X^p + \|y\|_X^p.$$

If $T : X \to X$ is bounded, then, for every $s \ge 0$,

$$\|sT - \mathrm{Id}\|_{X\to X}^p \le 1 + s^p \|T\|_{X\to X}^p.$$

Consequently $\limsup_{s\downarrow 0} \|sT - \mathrm{Id}\| \le 1$.

Proof. For every $x \in X$,

$$\|(sT - \mathrm{Id})x\|_X^p \le \|x\|_X^p + s^p \|Tx\|_X^p \le (1 + s^p \|T\|^p)\|x\|_X^p.$$

Take the supremum over nonzero x, then let s decrease to zero. □

4 The solution

Theorem 2. *Under the hypotheses of Open Problem 2 in arXiv:2601.01821,*

$$\boxed{\varepsilon_p(A; r, R) = 1 \qquad \text{for every } 0 < p \le 1.}$$

Proof. Fix $p \in (0, 1]$. First choose a nonzero radial profile $\eta_0 \in C_c^\infty((r, R))$, and define

$$\widehat{\psi}_0(\xi) = \eta_0(|\xi|).$$

This is the same kind of base generator constructed in Step 2 of the source's Hilbert-endpoint proof. Its Fourier transform is smooth, radial, and compactly supported away from the origin. Hence ψ_0 is Schwartz, has vanishing moments of every order, and belongs to $\mathfrak{R}(A; r, R)$.

The corresponding diagonal frame operator is bounded on H^p_A. Here is the standard multiplier justification. Under the source's no-aliasing support hypothesis, it is the Fourier multiplier with symbol

$$D_0(\xi) = \sum_{j\in\mathbb{Z}} \left|\widehat{\psi}_0((A^*)^{-j}\xi)\right|^2.$$

Finite overlap of the dilated annular supports makes the sum locally finite. It is smooth on $\mathbb{R}^n \setminus \{0\}$, is invariant under the dilation A^*, and has finite anisotropic Mihlin seminorms of every required order. The anisotropic Hardy multiplier theorem, equivalently the bounded analysis–synthesis theory for smooth wavelet molecules, therefore gives

$$M_p := \|U_{\psi_0,\psi_0}\|_{H^p_A \to H^p_A} < \infty \qquad [2, \text{Ch. 1, Sec. 9}].$$

For each real $t > 0$, put $\psi_t = t\psi_0$. Positive scalar multiplication preserves radiality, Fourier support, moments, and decay, so $\psi_t \in \mathfrak{R}(A; r, R)$. This closure is also used explicitly in Step 3 of the source's Hilbert-endpoint proof. If the word "dual" in the standing admissibility assumption is taken as part of auxiliary pair data, an accompanying dual ϕ_0 is simply replaced by $t^{-1}\phi_0$.

Since t is real and positive, the definition of the frame operator gives

$$U_{\psi_t,\psi_t} = t^2 U_{\psi_0,\psi_0}.$$

The anisotropic Hardy quasi-norm satisfies the p-power triangle inequality used in the source's proof of its diagonal lower bound. Applying Lemma 1 with $s = t^2$ yields

$$\|U_{\psi_t,\psi_t} - \mathrm{Id}\|_{H^p_A \to H^p_A} \le (1 + t^{2p} M_p^p)^{1/p}.$$

Taking the infimum over admissible generators and then letting $t \downarrow 0$ gives $\varepsilon_p(A; r, R) \le 1$. Theorem 1 of the source gives the reverse inequality $\varepsilon_p(A; r, R) \ge 1$. Equality follows. □

5 Verification, limitations, and novelty

The proof has three independently checkable inputs: the source lower bound; boundedness of one smooth annular multiplier on H^p_A; and the elementary cone/quadratic identities. It uses no numerical computation. In particular, the upper estimate uses the exact p-power triangle inequality, so it does not accidentally invoke local convexity when $p < 1$.

The theorem determines an infimum. If the intended generator class excludes the zero function, the damping sequence need not attain that infimum. The source asks only for its value. The paper's separate Open Problem 3 asks whether the lower bound survives after radiality is removed; scalar damping does not address that lower-bound question, so it remains open.

A bounded primary-source search through 12 August 2026 used arXiv:2601.01821, the exact current title, the phrases *saturation infimum*, *anisotropic Hardy wavelet*, and scalar rescaling/damping, together with the notation ε_p. It found the source's current arXiv version and no later primary-source resolution. Novelty confidence is moderate; specialist review is recommended before any priority claim. The most important review point is the application of the standard anisotropic multiplier theorem to the displayed smooth dilation-invariant symbol.

References

[1] K.-C. Wang, *An Anisotropic Balian–Low Phenomenon: Geometric Obstructions to Wavelet Frames*, arXiv:2601.01821v2 (2026).

[2] M. Bownik, *Anisotropic Hardy Spaces and Wavelets*, Memoirs of the American Mathematical Society 164 (2003).